Proof of $BB_{SK}(7) = 41$

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This note proves that the maximum normal-form size produced by any terminating SK-combinator expression of length 7 is 41. The argument combines exhaustive enumeration of all length-7 terms with a structural non-termination proof for the only remaining unresolved candidates.

1 Exhaustive Space Enumeration and Filtering

To establish the exact value of $BB_{SK}(7) = 41$, we first execute a two-phase filtering process: exhaustive state-space enumeration followed by bounded normal-order execution.

1.1 Combinator State Space Enumeration

A combinator term of length n is defined as a binary expression tree containing exactly n leaf nodes, where each leaf node is an element of the alphabet $\Sigma = \{S, K\}$. The structure of the expression is determined by the arrangement of application nodes.

The total number of unlabeled strictly binary trees with n leaves is given by the $(n-1)$-th Catalan number, C_{n-1}. For each such tree structure, every leaf node has two independent choices (S or K). Therefore, the total size of the search space for terms of length n is

$$\text{TotalTerms}(n) = 2^n C_{n-1}.$$

For $n = 7$, the sixth Catalan number is 132. The total number of valid combinator terms of length 7 is therefore

$$2^7 \cdot 132 = 16896.$$

1.2 Bounded Normal-Order Filtering

A decider program evaluates all 16896 terms using leftmost-outermost (normal-order) reduction for at most 15 steps. A reference implementation of the enumeration and filtering strategy is available in the accompanying GitHub repository: `https://github.com/b00ne/bb-sk-7`. The terms are partitioned into two mutually exclusive categories:

- Halting terms ($\leq$ 15 steps): terms that successfully reduce to a normal form (a state containing no further reducible expressions, or redexes) within 15 steps. The maximal combinator-count normal form observed among all terminating terms in this space is 41 combinators, achieved by the normal form of the term $SSS(S(SS))S$.
- Unresolved terms: terms that have not reduced to a normal form after 15 reduction steps.

Out of the 16896 total candidates, exactly two remain unresolved after 15 steps of reduction:

$$T_1 = S(SS)SSSS,$$
$$T_2 = SSS(SS)SS.$$

1.3 Equivalence via a Single Reduction Step

Notational note: throughout this proof, the reduction arrow $\xrightarrow{n}$ and step count n refer strictly to standard normal-order (leftmost-outermost) reduction, unless otherwise specified.

It is immediate that

$$T_1 \xrightarrow{1} SSS(SS)SS = T_2.$$

Because T_1 reduces to T_2 in exactly one normal-order step, their termination behaviors are identical:

$$T_1 \text{ is divergent} \iff T_2 \text{ is divergent.}$$

Consequently, proving the termination behavior of $T_1 = S(SS)SSSS$ simultaneously settles the outcome for both unresolved candidates.

1.4 Proof Strategy

To prove divergence of $T_1 = S(SS)SSSS$, we construct an infinite family of expressions M_n such that each instance reduces into a larger structure containing the next instance. By establishing that the initial term $S(SS)SSSS$ can be reduced to an expression of the form FM_0 (where F and M_0 are explicitly defined combinator terms), and that each M_n generates an infinite reduction chain, we demonstrate that $S(SS)SSSS$ itself admits an infinite reduction path, proving divergence.

2 Proof of Non-Termination for $S(SS)SSSS$

2.1 Preliminaries: contexts and reduction

Contexts. A *context* $\mathcal{C}[\cdot]$ is a combinator expression with a single distinguished hole. We write $\mathcal{C}[t]$ for the result of replacing the hole by t.

Reduction relation (definition). Let $\mathcal{R}$ be the set of primitive rewrite rules (here the S-rule). Define the one-step reduction relation $\to$ to be the *smallest* relation containing $\mathcal{R}$ and *closed under contexts*, i.e.:

$$\text{if } r \in \mathcal{R} \text{ and } r:\ \ell \to r',\ \text{then } \ell \to r',$$

and

$$\text{if } t \to t' \text{ then } \mathcal{C}[t] \to \mathcal{C}[t'] \text{ for every context } \mathcal{C}[\cdot].$$

Write $\to^*$ for the reflexive transitive closure of $\to$.

Remark 1 (Context closure). By definition of the reduction relation (contextual closure of the primitive rules), if $A \to^* B$ then for every context $\mathcal{C}[\cdot]$ we have

$$\mathcal{C}[A] \to^* \mathcal{C}[B].$$

This contextual closure is standard; see [1, 2].

Lemma 1 (Persistence of divergence in pure S)**.** *Let E be a pure S-combinator expression (no K or other erasing combinators). If E contains a subterm T that has no normal form (i.e., T is divergent), then E has no normal form.*

Proof. By definition of $\to$ (context closure), every reduction step inside T is also a reduction step inside $E = \mathcal{C}[T]$. Since T has an infinite reduction sequence, so does $\mathcal{C}[T]$. Equivalently, observe that the S-rule

$$Sxyz \to xz(yz)$$

does not discard any of its three arguments; hence no primitive rule erases a subterm. Therefore a divergent subterm cannot be eliminated by any finite sequence of reductions, and the whole term has no normal form. □

2.2 Preliminary Definitions

Let x be a fixed dummy variable. Define

$$\begin{aligned} F &= S(S(SS)S)S, \\ D &= S(S(S(S(SS)F))), \\ M_0 &= Dx, \\ M_n &= SM_{n-1}\big(x(SM_{n-1})\big) \quad \text{for } n \geq 1. \end{aligned}$$

2.3 Inductive Cycle

We establish that the family of expressions M_n forms an infinite reduction cycle by breaking the transformation down into three phases.

Notation: evaluation contexts. Throughout this section, each $\mathcal{C}_i[\cdot]$ denotes a fixed evaluation context (a combinator expression with a single distinguished hole, as in Section 2.1). The exact shapes of these contexts are unimportant; the argument applies structurally regardless of their internal structure.

Phase A: Expansion of FM_n. For any n, the application of FM_n unfolds to reveal a self-application pattern:

$$FM_n \xrightarrow{*} \mathcal{C}_0\big[M_n(SM_n)\big].$$

By expanding the definition of F and applying it to an arbitrary term t, a direct three-step normal-order reduction gives

$$Ft \xrightarrow{3} \mathcal{C}_0\big[t(St)\big].$$

(See Appendix A for the three-step trace of Ft.)

Substituting M_n for t produces the claimed context.

Phase B: Reduction to the base case. Any expression M_n applied to SM_n reduces through successive applications of M_n to the base case:

$$M_n(SM_n) \xrightarrow{*} \mathcal{C}_1\big[M_0(SM_n)\big].$$

By definition,

$$M_n = SM_{n-1}\big(x(SM_{n-1})\big).$$

Applying this definition to an arbitrary term t yields

$$M_n t \to M_{n-1}t\big(x(SM_{n-1})t\big).$$

Every application of $M_n t$ thus structurally reduces to an application of $M_{n-1}t$, and induction on n shows the expression unrolls until it reaches the base term M_0. Substituting SM_n for t gives the claimed target context.

Phase C: Index advancement. The base term M_0 increments the family index when it is applied to SM_n:

$$M_0(SM_n) \xrightarrow{*} \mathcal{C}_2[FM_{n+1}].$$

Using $M_0 t = Dxt$, after four reduction steps the base application produces the term $F(t(xt))$:

$$Dxt \xrightarrow{4} \mathcal{C}_2[F(t(xt))].$$

(See Appendix A for the four-step trace of Dxt.)

Substituting SM_n for t gives

$$M_0(SM_n) \xrightarrow{*} \mathcal{C}_2\big[F\big(SM_n(x(SM_n))\big)\big].$$

By the definition of the inductive family, the subterm $SM_n(x(SM_n))$ is exactly M_{n+1}, so the reduction simplifies to

$$M_0(SM_n) \xrightarrow{4} \mathcal{C}_2[FM_{n+1}].$$

Lemma 2. *For every $n \geq 0$, the term FM_n reduces, up to surrounding context, to a term of the form FM_{n+1}; that is, there exists a context $\mathcal{C}$ such that*

$$FM_n \;\to^*\; \mathcal{C}[FM_{n+1}].$$

Proof. Compose the three phases: Phase A unfolds FM_n to $\mathcal{C}_0[M_n(SM_n)]$, Phase B unrolls $M_n(SM_n)$ down to $\mathcal{C}_1[M_0(SM_n)]$, and Phase C advances the index to $\mathcal{C}_2[FM_{n+1}]$. Each phase is structural and independent of n, so the composition yields the claim. □

By repeated application of Lemma 2, the family produces an infinite sequence of index increments

$$\mathcal{C}_a[FM_0] \xrightarrow{*} \mathcal{C}_b[FM_1] \xrightarrow{*} \mathcal{C}_c[FM_2] \xrightarrow{*} \cdots,$$

showing that the subterm FM_0 admits an infinite reduction path.

2.4 Initialization and Main Theorem

Theorem 1. *The pure S-combinator term $S(SS)SSSS$ is divergent (has no normal form).*

Proof. Begin by executing the first nine normal-order reduction steps of the initial term:

$$S(SS)SSSS \xrightarrow{9} S(S(SS)F)\big(F(S(SS)F)\big).$$

(See Appendix A for the complete trace.)

Write the left-hand subterm as $\mathcal{L} = S(S(SS)F)$ and the right-hand subterm as $\mathcal{R} = F(S(SS)F)$. For an S combinator to form a redex, it must have at least three arguments. In the global expression $\mathcal{LR}$, the head combinator $\mathcal{L}$ has only one argument, so top-level application cannot reduce. Furthermore, the internal subterm $S(SS)F$ has only two arguments and is therefore inert.

Because the left branch $\mathcal{L}$ is in normal form and cannot be reduced, normal-order reduction is forced to evaluate only the right-hand subterm. Empirical tracking of $\mathcal{R}$ shows that after exactly 62 evaluation steps, the expression expands to

$$F(S(SS)F) \xrightarrow{62} \mathcal{C}_a[F(DD)].$$

The first four steps of this reduction are shown in Appendix A.

The subterm DD represents the base term $M_0 = Dx$ with the dummy variable x instantiated as D. Since the behavior of the inductive cycle does not depend on the specific choice of x, we may write this term as M_0, yielding

$$F(S(SS)F) \xrightarrow{62} \mathcal{C}_a[FM_0].$$

We have proven inductively that the subterm FM_0 possesses an infinite reduction path. By Lemma 1, because the pure S-calculus cannot erase subexpressions, the existence of this infinite reduction path within $\mathcal{C}_a[FM_0]$ guarantees that the parent expression $\mathcal{R}$ has no normal form (is divergent). Consequently, the initial term $S(SS)SSSS$ is divergent. □

3 Conclusion

By coupling the exhaustive filtering with the structural induction proof showing that T_1 unrolls into an infinitely growing cycle

$$\mathcal{C}_a[FM_0] \xrightarrow{*} \mathcal{C}_b[FM_1] \xrightarrow{*} \mathcal{C}_c[FM_2] \xrightarrow{*} \cdots,$$

we prove that the only two unresolved terms in the length-7 term space are divergent. Because all other candidate terms of length 7 have been verified to halt in ≤ 15 steps, the term $SSS(S(SS))S$, which reduces to a term of length 41, is definitively the term with the maximum normal form size. Given that $BB_{SK}(n)$ is defined as the maximum term length of a normal form generated by a halting term of length n, and the normal form of $SSS(S(SS))S$ contains exactly 41 combinators, the seventh BB_{SK} value is established:

$$BB_{SK}(7) = 41.$$

References

[1] J. R. Hindley and J. P. Seldin, *Introduction to Combinators and λ-Calculus*, Cambridge University Press, 1986.

[2] Terese (ed.), *Term Rewriting Systems*, Cambridge Tracts in Theoretical Computer Science, Cambridge University Press, 2003.

A Representative Reduction Traces

This appendix provides step-by-step normal-order reduction traces for the key terms cited in the proof. Each reduction is presented as an ordered sequence of terms showing the successive states of the expression.

Trace 1: Ft (expansion of F)

```
1  Ft
2  S(SS)St(St)
3  SSt(St)(St)
4  S(St)(t(St))(St)
```

Legend: The reduction of Ft shows how the definition $F = S(S(SS)S)S$ unfolds under application to an arbitrary term t, yielding the self-application pattern $t(St)$.

Trace 2: Dxt (expansion of D)

```
1  Dxt
2  S(S(S(SS)F))t(xt)
3  S(S(SS)F)(xt)(t(xt))
4  S(SS)F(t(xt))(xt(t(xt)))
5  SS(t(xt))(F(t(xt)))(xt(t(xt)))
```

Legend: The reduction of Dxt shows the expansion of the base term constructor, ultimately producing the form $F(t(xt))$ required for the inductive cycle.

Trace 3: $S(SS)SSSS$ (initialization)

Normal-order reduction of the initial term $S(SS)SSSS$ for 9 steps:

```
1  S(SS)SSSS
2  SSS(SS)SS
3  S(SS)(S(SS))SS
4  SSS(S(SS)S)S
5  S(S(SS)S)(S(S(SS)S))S
6  S(SS)SSF
7  SSS(SS)F
8  S(SS)(S(SS))F
9  SSF(S(SS)F)
10 S(S(SS)F)(F(S(SS)F))
```

Legend: The reduction of $S(SS)SSSS$ shows how the initial term unfolds in 9 steps to reach the form $S(S(SS)F)(F(S(SS)F))$, which splits into the left-normal-form component and the right-hand term that expands to $\mathcal{C}_a[F(DD)]$.

Trace 4: $F(S(SS)F)$ (excerpt)

The complete 62-step normal-order reduction of $F(S(SS)F)$ is too long to include in this document. The first 4 steps are shown below:

```
1  F(S(SS)F)
2  S(SS)S(S(SS)F)(S(S(SS)F))
```

```
3  SS(S(SS)F)(S(S(SS)F))(S(S(SS)F))
4  S(S(S(SS)F))(S(SS)F(S(S(SS)F)))(S(S(SS)F))
5  S(S(SS)F)(S(S(SS)F))(S(SS)F(S(S(SS)F))(S(S(SS)F)))
```

Full trace availability: Readers interested in verifying the complete 62-step reduction may consult the repository trace file `full-62-step-trace.txt` at the GitHub repository `https://github.com/b00ne/bb-sk-7`.
Commit `5082c89bccfd3dc825dbcdbb8eb6d7ea74e2dff4` contains the full step-by-step sequence.